

BOERHAAVE'S NATIVE UREA AND THE ART OF FRACTIONAL CRYSTALLIZATION

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Summary

The history of one major chemical in science, agriculture and technology began with Herman Boerhaave's isolation of native urea in the early 18th century. It has been overlooked up to now that this was the first instance of a special kind of fractional crystallization being utilized, albeit unwittingly, for the separation of two compounds (urea and sodium chloride). A detailed quantitative explanation, therefore, is provided in the present article. Boerhaave's work is well suited to be included in today's teaching of chemistry in the category of "historic experiment redone," not least by virtue of the largely innocuous nature of the substance involved (urea). A pertinent trial experiment has, therefore, been carried out, requiring virtually no additional chemicals and calling for only the most basic equipment, but yielding a most beautiful result: a surprisingly abundant crop of native urea crystals. In addition a closer look at the historical literature has revealed that the first description of urea obtained by Boerhaave's process was published—earlier than assumed previously—almost exactly 300 years ago, in 1724. In conclusion the merits of reading the primary chemical literature and carrying out experimental work in the history of chemistry are briefly discussed.

Introduction

Urea has attracted special attention in the past and continues to do so at the present time, be it in science, agriculture or technology. It was the first organic compound to be synthesized (Friedrich Wöhler, 1828) (1), it is still incredibly important as a fertilizer in agriculture—the annual production amounting to roughly 200 Mt (megatonnes)—and it serves increasingly in technology as a scavenger of nitrogen oxides from Diesel engines in the SCR process (selective catalytic reduction). Regarding its history two misconceptions keep coming up. As the late Jack Dunitz and coworkers noted some time ago, Wöhler in his synthesis of urea in 1828 did *not* employ ammonium

cyanate as a starting material, contrary to what is widely believed (2). Ammonium cyanate was unknown back then and in fact Wöhler intended to prepare it for the first time by reacting either ammonium chloride solution with silver cyanate or ammonia solution with lead cyanate. In both reactions he obtained urea instead of the expected inorganic salt. Yet text books of organic chemistry today still tend to have it that Wöhler used ammonium cyanate, a notable exception being the "Vollhardt-Schore" which not only gets the history right but dedicates almost a whole page to a thorough discussion of urea (3).

The second misconception concerns the question of who in fact discovered urea. One may still find it stated that originally Hilaire-Marin Rouelle (1718-1779) in France obtained it in 1773 (4). But as Backer pointed out in 1943 in a paper written in a spirit of Dutch chemical patriotism it was Herman Boerhaave (1668-1738), the Dutch botanist, physician and chemist, who isolated urea in crystalline form from urine for the first time in the early 18th century (5). Kurzer and Sanderson, who give an excellent, well-documented account of the story, date the discovery to some time before 1727 (6). But actually the first mention in print dates back to 1724 as will be shown below. In fact it was Kurzer and Sanderson's account which directed our attention to Boerhaave's work, and the present article may, therefore, be regarded as an independent supplement to their paper on the early history of urea.

Why Urea?

The dissolved solids in an adult's urine collected over a period of 24 hours—approximately 1.5 L in volume—amount on average to (7):

28 g of urea ($\text{OC}(\text{NH}_2)_2$),

11 g of sodium chloride (NaCl),

5g of potassium sulfate (K_2SO_4),

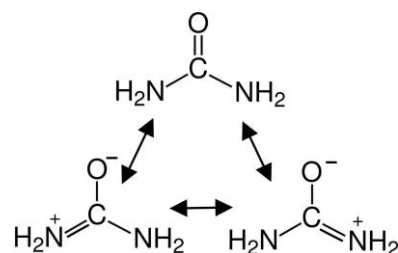
4 g of primary sodium phosphate (NaH_2PO_4)

and minor amounts of other organic and inorganic constituents (all values $\pm 25\%$). With respect to urea, the main constituent, given a world population of about 10 billion (ten times 10^9) this corresponds to some 100

Mt annually, easily one half of the aforementioned present-day industrial output! Obviously, being thus abundant, urea could not have remained hidden from chemical investigators for long, which may in part explain the very early discovery by Boerhaave. We, however, were intrigued by what seemed to be another important factor—previously overlooked—which contributed to Boerhaave’s success in isolating urea in pristine crystalline form for the first time, viz. a specific kind of *fractional crystallization* which he apparently unwittingly took advantage of. It enabled him efficiently to separate urea from sodium chloride, the second major constituent of urine besides water, as will be explained in a subsequent section.

While the primary aim of the present contribution is to call attention to this historical finding, urea is an object suited for historical study and chemical education in its own right. The epithet of a “glorious past” might be justified by a number of names—besides that of Boerhaave in The Netherlands—of prominent chemists associated with its history: Antoine François de Fourcroy (1755-1809) and Louis Nicolas Vauquelin (1763-1829)—these two also coined the name “urée,” or “urea”—and Joseph Louis Proust (1754-1826) in France; William Cruikshank (1745-1810 ?) and William Prout (1786-1850) in England; Jöns Jakob Berzelius (1779-1848) in Sweden; Friedrich Wöhler (1800-1882) in Germany. For details on the former of these, Kurzer and Sanderson’s inspiring paper should be consulted (6), for references on Wöhler’s original contribution the note by Dunitz et al. (2). But another important name should not be forgotten: Hermann Kolbe (1818-1884) in Germany. Together with one of his students—Alexander Basaroff—Kolbe in 1868 synthesized urea in an ingeniously simple way just by heating (at 130-140 °C in a sealed tube) dry ammonium carbamate which in turn is obtained by direct reaction (at room-temperature) of carbon dioxide and ammonia (8, 9)! This is essentially the way in which virtually all urea produced industrially is obtained today, by a continuous high-pressure process first employed on a large scale by the company BASF in Germany in 1914 (10). Although occasionally called *Basaroff-reaction* it would be more appropriate to name it *Kolbe-Basaroff-reaction* as it was on Kolbe’s instigation (and theoretical prediction) that Basaroff carried out the experimental investigation (8). It has been noted that this is possibly the largest artificial CO₂-capturing process in operation today (11). Ironically, however, almost all of the CO₂ thus fixated is again being released into the atmosphere since about 90% of the world’s urea production goes into soil fertilizers.

But urea seems now to be bound for a glorious future as well, for there is currently much research going on in the area of (electrical) energy storage materials. Here urea is a promising new candidate. In this connection also new synthetic routes to urea are being explored either by the help of novel catalysts or by direct electrocoreduction of carbon dioxide with, for instance, nitrate ions. These developments, however, are too recent and fast-evolving to be considered in the present contribution which is primarily concerned with history. Finally, for those engaged in chemical education, one may mention as an aside that the urea molecule illustrates the concept of resonance in analogy to the carbonate ion:



Thus a classical estimate yields about 20% double bond character for each of the two (nominally single) N–C bonds and only about 60% for the (nominally double) C–O bond (12).

Urea—300 Years on Record

Contrary to what Kurzer and Sanderson believed (6), the discovery of native urea was unequivocally made public in print (in Latin) as early as 1724 in what is now known as the first spurious (unauthorized) edition of Boerhaave’s lecture notes (13), presumably published in Leyden though Paris is given on its title page (14). Kurzer and Sanderson only refer to the later English edition of this work, published in London in 1727 (15), and therefore assume that it was then at the latest that urea could have become publicly known (even though they cite the original 1724 Latin edition too). Because of its historic significance and also to compare it with Boerhaave’s own improved version later on, the text is here reproduced in full along with its English (non-verbatim) translation (13, 15):

PROCESSUS OCTAVUS.

Sal Urinæ essentialis, seu sal essentialis Corporis Humani ex urinâ recenti inspissatâ.

§. 1. Si sumatur urina recens sanissimi hominis, inspissetur ad syrupi crassitiem, igne nunquam interrupto, filtretur ut oleositas in chartâ maneat, reponatur tunc in vase vitreo, in loco subterraneo frigido, fere per anni spatium, concrescet circa fundum, id est unius alteriusve digiti distantia ab

eo, crusta salina, crassa, fusca; effundatur prudenter omne quod supernatat liquidum, & sollicitè examinetur illa gleba salina, inferius enim hæret materia phosphori.

§. 2. Hæ glebæ salinæ purâ aquâ calidâ solvantur, filtrentur, inspissentur ad pelliculæ apparitionem, ponatur iterum materies in loco subterrano, crescent in fundum Crystalli, quæ separatim collectæ, salem humanum constituunt magis rectificatum, tunc quo plus hic processus repetitur, eo hic sal magis fit albus. Est summum diureticum & emmenagogum, quod natura novit, & diversus ab omni alio sale nostri corporis.

PROCESS LXXXVII [numbering differs from Latin original].

Shewing, that recent *URINE* will crystallize by inspissation, or afford an *essential salt*.

1. Evaporate a large quantity of the recent urine of a healthy man, to the consistence of a syrup, with one continued degree of heat; separate the oil from it by the filtre, and afterwards put it into a close vessel, and set it in a cool cellar, for the space of one year; it will shoot into a thick, brown, saline glebe, or crust, at the distance of an inch or two from the bottom of the vessel: then carefully pour off all the liquor that floats above, take out the mass of saline crystals, and lay the matter at the bottom thereof aside for phosphorus. These crystals, which are of a peculiar nature, being dissolv'd in pure warm water, filtred and evaporated to a pellicle, and set again to shoot in the cold, will again run into crystals *sui generis*, or a white purified salt, of a nature different from all other salts, and become the purer, the oftener the process is repeated; so that it may justly be call'd the *essential salt of urine*, or of the human body.

2. ... upon trial [I] find that it is of a wonderful diuretic and emmenagogic virtue ...

Note that conventional *recrystallization* (well-known to any chemistry major from practical organic chemistry) is here being recommended to purify the urea after its isolation (“... and become the purer, the oftener the process is repeated ...”). But this is not what makes Boerhaave’s process different from ordinary practice. Boerhaave’s method stands apart by a special kind of *fractional crystallization* (as will be explained in the next section) whose details, however, were omitted in the above protocol possibly due to a lack of grasp on the part of those attendants of Boerhaave’s lectures on whose notes this unauthorized text was based. In any case, in

view of the existence and exact dating of the Latin text (14), there can be no doubt that by 1724 urea had been prepared in pure crystalline form and the discovery disseminated that very year in a widely sold book—as Boerhaave himself admitted: “... this sorry Book met with too many Buyers...” (16). This then makes for a tri-centenary proper in 2024!

Salt and Urea: Part They Must

According to the earlier quoted concentrations of solutes the main constituents of urine are salt (sodium chloride, NaCl) and urea. Were one in possession of this knowledge *a priori* one could try and search for adequate means of separating these two. Boerhaave was not in possession of such *a priori* knowledge. It must, therefore, be regarded as yet another instance of that marvelous serendipity so often encountered in the history of science (or any other branch of human endeavor for that matter) that Boerhaave struck upon a very effective means of separation even though he could not have had more than an inkling—or hope—that something more interesting than ordinary table salt was waiting to be discovered—this “something” being urea. Boerhaave’s own authentic text, published in 1732 in Latin (17), appeared in 1735 as a close to verbatim English translation which is reproduced here (16):

[Vol. II, p 220] PROCESS XCVIII.

The Native Salt of Urine.

APPARATUS.

TAKE some very fresh well-concocted Urine of persons in perfect Health, put it presently into a very clean Vessel, and with an equable Heat of 200 degrees, evaporate it till you have reduc'd it to the consistence of fresh Cream, and whilst it continues thus hot strain it through a bag, that the tenacious Oil may in some measure be retain'd there, and separated from it; and the more accurately this is done, the better. Put a large quantity of this thick inspissated Liquor into a tall cylindrical glass Vessel, with a paper tied over it, and let it stand quiet in a cool place for the space of a Year. By this means, then, you will have a solid, hard, sub-pellucid, brown, saline Mass, fix'd all about the bottom of the Vessel; and over this a thick, black, pinguious Liquid, separated and rejected as it were from the concreted Salt. Decant this Liquor, take out the saline Mass, put it into another Vessel, pour some very cold Water upon it, and shake it about to free it from its oily Impurities, which may be done pretty easily, as it will not readily dissolve in cold Water. Keep this saline Matter under its proper title. If this is

dissolved in hot Water, and strained till the *Lixivium* becomes exceedingly limpid, and evaporated to a Pellicle in a very clean glass Vessel, then, if you set it by in a cold place, it will shoot into saline Glebes of a particular kind, that are perfectly distinct from every other salt. In their figure, and solidity, however, they come pretty near to the Crystals of Sugar. These are not fetid, nor alkaline, but very volatile. This is the native Salt of Urine.

[Vol. I, p 42] ... if for a good while you apply to it a degree of Heat, little greater than that of boiling Water, it will become intirely volatile.

According to this text, the means by which Boerhaave accomplished the separation of urea from sodium chloride (NaCl) consisted in filtering—*while still hot*—the suspension obtained by first evaporating the urine to a quite small volume (he used a bag to strain the solution through rather than a filter). Actually he was unaware of accomplishing this separation, for he does not mention the salt (NaCl) and assumes that it was only the “tenacious oil” which was being detained by the filter. With today’s knowledge, however, it is clear that the NaCl would have largely crystallized out due to the small volume to which the urine had been evaporated and would thus remain on the filter. Here the crucial point is that the solubility of NaCl is almost temperature-independent while that of urea increases enormously with rising temperature (Figure 1). Therefore urea remains in solution *as long as the mixture is hot* but after filtration will crystallize in the cold whereas NaCl will not. A repetition of the experiment corroborates this interpretation (see penultimate section below) for it is found that after filtering the hot concentrated urine from the precipitated sand-like NaCl (and other insoluble minor constituents) the dark-colored liquid remains perfectly clear in the cold except for the slowly growing distinctly shaped urea crystals. Thus one must marvel at the providential circumstance that the most abundant constituent accompanying urea in urine happens to exhibit a practically temperature-independent solubility allowing complete separation in a single step.

The next most abundant constituent on the list previously given, potassium sulfate (K_2SO_4), is the least soluble one of those named (18), especially at low temperature (and also less soluble than either potassium chloride, KCl, or sodium sulfate, Na_2SO_4 , see ref. 7, note). Were part of the K_2SO_4 to crystallize out in the cold one would, however, still be able to remove it afterwards by recrystallization of the urea as advised particularly by the unauthorized version of the text quoted previously. (We note in passing that the term

“recrystallization” is used by metallurgists quite differently to denote a transformation of grain structure in metals.)

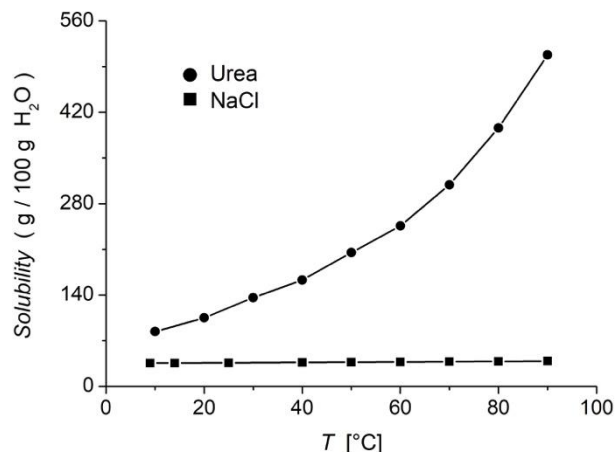


Figure 1. Solubility of Urea and sodium chloride (NaCl) as a function of temperature (values from refs. 18 and 19 respectively); the nearly constant solubility of NaCl enabled Boerhaave to separate it from urea in a single step (by hot filtration) even though he was unaware of it.

Comparing the unauthorized version with Boerhaave’s authentic text one sees that the most crucial detail escaped the authors of the former since they did not stress that the suspension be filtered while still hot. Much later Kurzer and Sanderson failed to notice this too, for they merely note that the “‘pirated’ edition of 1727 ... contains practically the same account on the isolation and properties of urea as the genuine textbook of 1732” (ref. 6, p 453).

That the true rationale behind the successful separation of the two compounds urea and NaCl was not known to Boerhaave himself may also be inferred from the fact that he thought it necessary to let the filtrate crystallize for the incredibly long period of an entire year. In reality a few hours suffice for the urea to start crystallizing out of the cold solution. Still Boerhaave employed the crucial hot filtration quite consciously as he makes clear by pointing out that “the more accurately this is done, the better.”

A second omission in the unauthorized edition concerns the temperature at which the fresh urine was to be evaporated. Boerhaave gives the explicit value of 200 degrees (Fahrenheit), or about 90 °C (assuming a round number). In retrospect it makes sense to use a reduced temperature rather than heating the solution to the boiling point since urea undergoes hydrolysis in water, a reaction which must be avoided and which strongly depends on temperature. In this connection a third detail missing in

the unauthorized edition is a mention of the fact that the largely evaporated urine attains “the consistence of fresh cream,” as Boerhaave puts it, which is due to the said slow hydrolysis of urea, yielding ammonia and carbon dioxide. These two gases upon liberation turn the concentrated urine into wet foam. This phenomenon too is reproducible by carrying out the experiment today. Incidentally this is best done in the open or under good ventilation as the smell of ammonia (and also that of stale urine) may be deemed unpleasant by some.

Much Ado about Nothing

Both Backer and afterwards Kurzer and Sanderson were concerned about the undesirable possibility of salt (NaCl) and urea crystallizing together as the known 1:1 addition compound (monohydrate) rather than each of them separately (5, 6). Backer, together with his coworker J. de Jonge as he says, carried out a number of trial experiments with solutions of varying concentration of the two solutes as well as with two samples of urine, the latter having been decolorized and deodorized by treatment with activated charcoal. Most often the two solutes crystallized separately. Only in two cases did the investigators obtain the binary complex. Looking again at their report one can see that this was only the case when the concentration of the two components was of a similar value (both near saturation) and crystallization occurred in the cold (room temperature). They, however, did not take notice of this correlation, for even though in Boerhaave’s experiment the latter conditions did *not* obtain, Backer and de Jonge still held co-crystallization to be a possibility (5). Kurzer and Sanderson afterwards tacitly adopted this view too (6). But both these couples of investigators could have spared themselves the worry. For in fact Boerhaave’s account of the experiment leaves no doubt that already the first batch of crystals obtained by his method could not have contained NaCl: re-dissolved in water and “evaporated to a pellicle,” if then set by “in a cold place” they would “shoot into saline glebes” once more, he writes. Had NaCl been present it would have crystallized out already *while the solution was being evaporated*—just as had happened while evaporating the fresh urine—instead of crystallizing only in the cold! (Note that in urine the concentration of NaCl relative to that of urea is even lower than it would have been in the 1:1 addition compound, yet crystallization sets in already during evaporation.) Attentive reading of old texts—if meticulously written, like the present one by Boerhaave—is thus seen to be no futile exercise.

Historic Experiment Redone

As noted in the foregoing section, Backer and J. de Jonge already carried out what one may call two proof-of-principle experiments with urine to ascertain that the isolation of urea by the method of Boerhaave is possible (4). But they deviated from Boerhaave’s instructions in several respects: (i) the urine was first decolorized and deodorized by means of activated charcoal, (ii) the urine was evaporated either on a water-bath (100 °C) or partly under reduced pressure and partly on the latter and (iii) the crystallization of the final product took place either at a lowered urea concentration (partly due to decomposition at the water-bath temperature)—leading to the aforementioned addition-compound—or artificially by “seeding” the solution with pure urea crystals. Therefore—but also to gain some out-of-the-ordinary hands-on experience of physical reality as an antidote to today’s obsession with virtual reality—Boerhaave’s experiment has been reproduced as follows.

Over a period of five days, the entire urine (amounting to 6.4 L) of a healthy adult was collected using a 0.5 L PET (polyethylene terephthalate) screw cap bottle, each portion being emptied at once into a 2 L stainless steel (aluminum kernel bottom) pot placed on a temperature-controlled hot-plate (without stirring) set to a temperature of 90 °C. This plate-temperature setting was chosen so that the urine was being evaporated at very nearly the rate it came in (the actual temperature of the liquid was around 80 °C throughout). After the last portion had been added, the heating was continued for about 24 hours more until the liquid acquired, in Boerhaave’s words (see above), “the consistence of fresh cream,” meaning that it formed finely dispersed wet foam. Letting the contents temporarily cool to ambient temperature overnight resulted in needle-shaped crystals growing out of the dark brown suspension now free of foam. These urea crystals confirmed that the evaporation had progressed sufficiently far. Upon reheating the suspension to the same temperature as before the crystals re-dissolved and the foam formation, obviously due to slow evolution of gases, resumed. The total residue then weighed 336 g and was poured in three successive portions onto a doubly folded 18 cm (diameter) plain round paper filter in a glass filter funnel preheated by letting boiling hot water run through the filter just before pouring in the hot urine suspension and placed on a 300 mL wide-neck Erlenmeyer flask (Figure 2).



Figure 2. Preparing well-crystallized urea from urine: fresh portions of urine (a) continually added into the pot on the hot-plate (b) for five days and evaporated to “the consistence of fresh cream” (Boerhaave), then filtered hot by employing a hot-air pistol (c), yielding upon cooling a copious crop of quite beautiful centimeter-long urea crystals (d) shown close-up in the inset.

After 4 hours when about 100 mL of filtrate had been collected and the filter funnel had long since cooled down to ambient temperature the filtration became exceedingly slow. Therefore the funnel was heated by means of a hot air pistol (“hair dryer”) from the sides while stirring the half sandy (NaCl), half pasty (from other urine constituents) filter residue with a glass rod, taking care not to inadvertently perforate the filter paper (the glass rod had a rounded end). This was repeated three times over a period of 24 hours, yielding a total volume of the clear, dark red-brown, almost black filtrate of nearly 150 mL weighing 203 g. Therein crystallization of centimeter-long transparent needles typical of urea had started already after some hours in the course of the filtration. But the crystal growth being slow, it may be assumed that this leads to but a minor loss due to some crystallization on the filter. Afterwards the filtrate was

kept in a refrigerator (7 °C; flask opening covered with polyethylene film) to promote further crystallization.

Urea crystals are known to adopt the shape of long quadratic prisms (“needles”) with poorly developed octahedron-type end-faces. In contrast, NaCl crystallizes in cubes or (when urea is present in the mother liquor) octahedra, and the urea-NaCl addition compound in short rhombic prisms which almost appear like flat hexagons. Crystallographic descriptions of the first and last, with illustrations, were provided by Werther almost two centuries ago (20). Thus the shape of urea crystals (“needles”) stands apart and is easily recognized (Figure 2, inset).

As indicated earlier the evaporation of urine is accompanied by a penetrating smell due in part to ammonia being evolved (along with carbon dioxide) as a result of slow hydrolysis of urea, clearly observable towards the end by the formation of a slowly rising wet foam even well below the boiling point of the solution. (As also remarked previously this is the reason why a moderate temperature during the evaporation is advantageous since the hydrolysis is strongly temperature dependent.) Other constituents of stale urine—even though kept at about 80 °C throughout so that no fermentation, or “organic” decomposition, should occur—contribute to the smell. Yet, although this experiment was carried out without special precautions or additional ventilation it had no negative after-effects on-site, as the smell disappears soon after the evaporation is stopped. Incidentally one of the possible solid-state products—the least volatile one—from the reaction of ammonia, carbon dioxide and air moisture would be ammonium bicarbonate, or hydrogen carbonate (NH_4HCO_3 , also known as “ABC”), which is one kind of baking powder commonly used at Christmas time.

Conclusion

Two lessons may be learnt from the foregoing: (i) chemical literature deserves to be read in its original form by consulting the *primary sources* besides secondary literature, and (ii) this reading should be done *conscientiously*, i.e., one should always painstakingly ask oneself about the meaning of what one is reading. Had this been generally adhered to then Wöhler’s synthesis would not be misrepresented in so many textbooks still today. Also the discovery of urea would then no longer be ascribed to Rouelle but to Boerhaave. And as has been shown in the present contribution the real physical reasons behind Boerhaave’s successful isolation of urea—suppression of hydrolysis and separation of sodium chloride (NaCl)—could not have escaped notice

until now. Not even the minor uncertainty about whether Boerhaave had really isolated urea as such or merely as its NaCl addition compound could then have arisen. Finally the true tri-centenary of the discovery of urea could not then have been missed in 2024. It is hoped that by having dwelled at length throughout the preceding three sections on all the minutiae of Boerhaave's process of preparing native urea it has become apparent what richness of experience, insight and information may be hidden in a particular historical text even to an extent not suspected at first sight.

But one may take a broader perspective in connection with today's general reading culture. As has been noted repeatedly for a number of years the tendency nowadays is not to *read* any more but just to *scan* texts. Readers seem to find it ever more difficult to completely immerse themselves in a book or a written paper. Neuroscientists and those engaged in educational research warn that incessant use of digital media from early age on may change the neuronal architecture of the developing brain, which in turn could impair people's ability to think and make proper judgments, with unforeseeable consequences for society in the long run (21). This may already be noticeable in the chemical literature, an area closer to our hearts perhaps: in one particular case (possibly but the peak of the iceberg) it was clearly demonstrated not long ago that sometimes neither editors nor referees of chemistry journals really (conscientiously) read the papers they publish any more (22).

Finally, when teaching—or studying—chemistry, it should be borne in mind that redoing historical experiments with one's own hands is of unique value. It is as if another dimension were added to the “reliving” of history, a dimension inaccessible to the non-experimental disciplines of the humanities. In the field of study dedicated to the role of museums in society the term “aura of the original” is used to designate the peculiar effect which one experiences in the actual (physical) presence of historical objects (23). This may sound mystical at first but many would agree that historical objects do indeed exert some specific

influence. Something similar may be happening when one performs authentic historical experiments. An intriguing aspect of this is that the “aura of the original” can be evoked in chemistry by generic chemical substances, i.e. without the need of genuinely historical “objects.”

One's own personal experience of chemical substances in the real world through experiments, therefore, is seen to be of special importance, a sentiment Boerhaave would have been the first to subscribe to, judging from two quotations from his biography written by the notable 18th century critic and author Samuel Johnson (1709-1784) in 1739, notwithstanding their characteristic cultural and stylistic “patina” (24):

[p 115] When he [Boerhaave] laid down his Office of Governor of the University [of Leyden] in 1715, he made an Oration upon the Subject of *attaining to Certainty in natural Philosophy*; in which he declares, in the strongest Terms, in favour of Experimental Knowledge, and reflects with just Severity upon those arrogant Philosophers, who are too easily disgusted with the slow Methods of obtaining true Notions by frequent Experiments ...

[p 175] He examined Systems by Experiments, and formed Experiments into Systems. He neither neglected the Observations of others, nor blindly submitted to celebrated Names. He neither thought so highly of himself as to imagine he could receive no Light from Books, nor so meanly as to believe he could discover nothing but what was to be learned from them. He examined the Observations of other Men, but trusted only to his own.

Acknowledgment

In memoriam Rev. Victor J. Gustitus (1943-2024) who introduced the author not only to Dr. Johnson, the early Boerhaave biographer, but to the vast realm of English and American literature.

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About the Author

Marek Petrik, a materials scientist by training, takes also an active interest in the history of chemistry and in post-publication peer-review (see for instance ref. 22).